FRANCK PORTIER
2016-2017

M1 MACRO

Group B

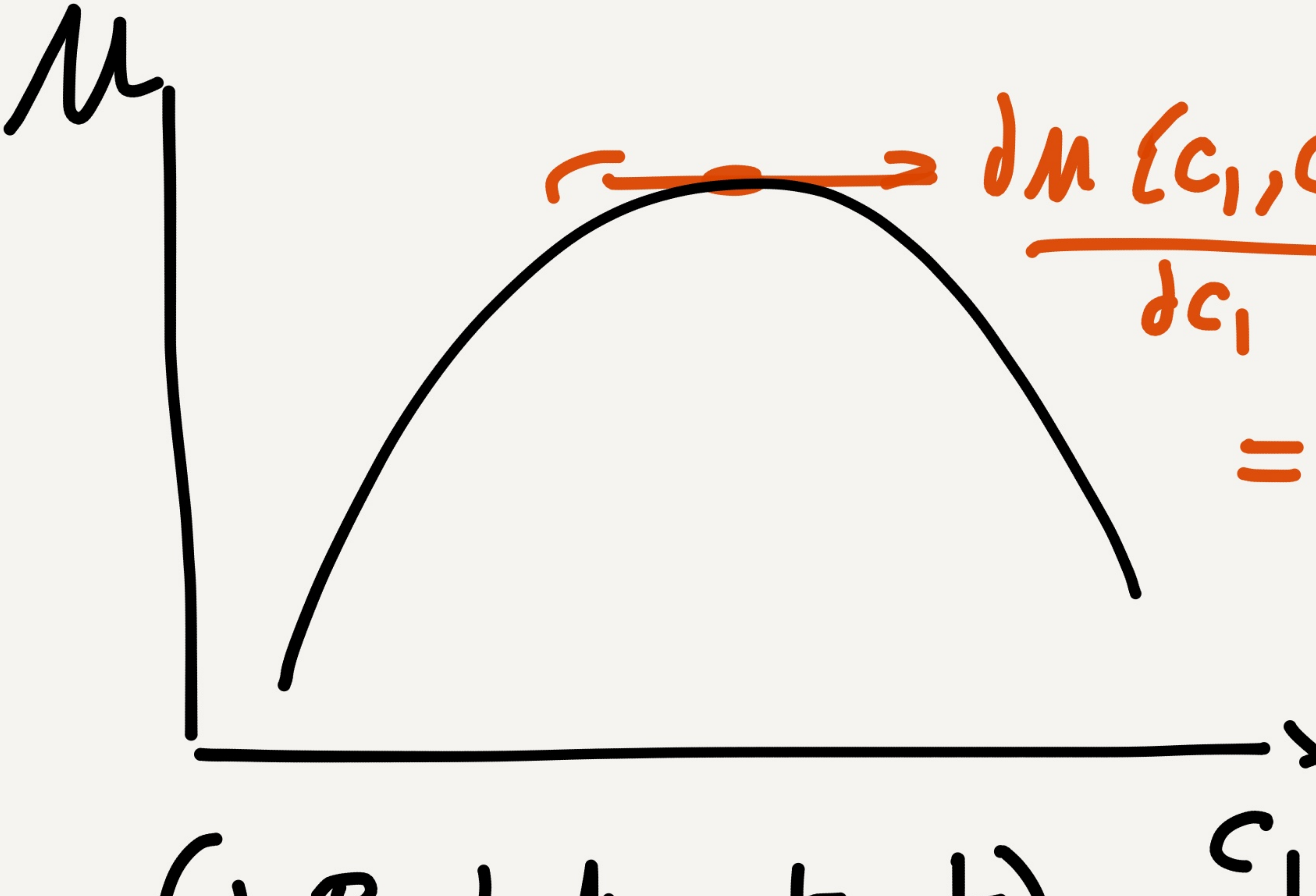
Session 2

These notes are written during class. They are posted "for the record" and may contant mistakes. They are not intended to be read by anyone that did not attend the class.

$$\left[\begin{array}{l} \max \log c_1 + \beta \log c_2 \\ \text{s.t.} \quad p_1 c_1 + p_2 c_2 \underset{=}{\le} p_1 y_1 + p_2 y_2 \end{array}\right.$$

$$\Leftrightarrow \quad \underbrace{c_2}_{c_2(c_1)} = \frac{p_1 y_1 + p_2 y_2 - p_1 c_1}{p_2}$$

$$u = \log \underline{c_1} + \beta \log \left(\frac{p_1 y_1 + p_2 y_2 - p_1 \underline{c_1}}{p_2} \right)$$



$$\frac{du(c_1, c_2(c_1))}{dc_1} = 0$$

(+ Budget constraint)

FOC

$$\frac{\partial u}{\partial c_1} = \frac{1}{c_1} + \beta\left(-\frac{p_1}{p_2}\right)\frac{1}{c_2} = 0$$

$$\Leftrightarrow \boxed{\frac{\beta\, 1/c_2}{1/c_1} = \frac{p_2}{p_1}}$$

where we have used: $\left[\log(f(x))\right]' = f'(x) \times \frac{1}{f(x)}$

$$\frac{\beta / c_2}{1 / c_1} = \frac{p_2}{p_1} \iff c_2 = \frac{p_1}{p_2} \beta c_1$$

plug it back in the BC

$$p_1 c_1 + p_2 \left[\frac{p_1}{p_2} \beta c_1 \right] = p_1 y_1 + p_2 y_2$$

$$p_1 (1+\beta) c_1 = p_1 y_1 + p_2 y_2$$

$$c_1 = \frac{1}{1+\beta} \left[y_1 + \frac{p_2}{p_1} y_2 \right]$$

SOLUTION

Units

C_1 : # of apples in period 1

Y_1 : ————————

C_2 : ———————— 2

Y_2 : ———————— 2

P_1 : €/apple of period 1

P_2 : €/ ———— 2

$$\frac{P_2}{P_1} : \frac{€/\text{ap.}2}{€/\text{ap.}1} : \frac{\text{app.}1}{\text{app.}2}$$

$$c_2 = \frac{p_1}{p_2}\beta c_1 = \frac{p_1}{p_2}\beta\left[\frac{1}{1+\beta}\left(y_1 + \frac{p_2}{p_1}y_2\right)\right]$$

$$c_2 = \frac{\beta}{1+\beta}\left(\frac{p_1}{p_2}y_1 + y_2\right)$$

For prices (p_1, p_2) (arbitrary),

Supply: y_1

y_2

Demand: $c_1 = \frac{1}{1+\beta}\left(y_1 + \frac{p_2}{p_1} y_2\right)$

$c_2 = \frac{\beta}{1+\beta}\left(\frac{p_1}{p_2} y_1 + y_2\right)$

- Let's find prices such that $D = S$ on all markets.

2 prices P_1, P_2

2 markets $c_1 = y_1$, $c_2 = y_2$

Let's solve first $c_1 = y_1$

$$c_1 = \frac{1}{1+\beta}\left(y_1 + \frac{p_2}{p_1} y_2\right) = y_1$$

$$\left(\frac{1}{1+\beta} - 1\right) y_1 = -\frac{p_2}{p_1} y_2 \times \frac{1}{1+\beta}$$

$$\boxed{\beta \frac{y_1}{y_2} = \frac{p_2}{p_1}}$$

Eq. on good 2 market

$$C_2 = Y_2$$

$$\frac{\beta}{1+\beta}\left(\frac{p_1 Y_1}{p_2} + Y_2\right) = Y_2$$

Same equation

$$\frac{\beta}{1+\beta}\frac{p_1}{p_2}Y_1 = \frac{1}{1+\beta}Y_2$$

$$\frac{p_1}{p_2} = \frac{1}{\beta}\frac{Y_2}{Y_1} \Longleftrightarrow \boxed{\frac{p_2}{p_1} = \beta\frac{Y_1}{Y_2}}$$

WALRAS LAW

In an economy in which agents satisfy their budget constraints, If $n-1$ markets clear, then the nth market also clears

Walras Law in this model

BC $\quad p_1 c_1 + p_2 c_2 = p_1 y_1 + p_2 y_2$

equil. on good 1 market: $c_1 = y_1$

BC $\quad \cancel{p_1 y_1} + p_2 c_2 = \cancel{p_1 y_1} + p_2 y_2$

$c_2 = y_2 \Rightarrow$ good 2 market clears

numéraire : a good whose price is arbitrarily set to $\underline{1}$.

(often money)

→ Assume $p_1 = \underline{1}$

then $\boxed{p_2 = \beta \frac{y_1}{y_2}}$ **Eq. PRICE**

- Check that for $p_1 = 1$

$p_2 = \beta \dfrac{y_1}{y_2}$,

$c_1 = y_1$ and $c_2 = y_2$